用多路径

Snapdrive for Unix

NetApp
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目□

□用多路径 1

□用多路径

要在 SnapDrive for UNIX 中使用多路径，必□完成包括安装和□□ HBA 在内的□□。如果□使用的是非 NetApp 多路径解决方案，□必□从 HBA 供□商的网站下□□件和□用程序包。

□的系□必□□行当前版本的 FC 和 iSCSI 系□□件。

"NetApp 互操作性表工具"

□□

1. 在安装相□的 Host Utilities □件之前，□先安装支持的 HBA 。

   □用于 UNIX 的 SnapDrive 支持多供□商 HBA 。

2. □□ HBA 服□。

   如果 HBA 服□未□行，□在□行 SnapDrive for UNIX 命令□会□示以下□□消息，例如 SnapDrive storage create` 和 SnapDrive config prepare LUNs` ：

   ```
   0001-876 Admin error: HBA assistant not found
   ```

   □必□□□所需数量的路径是否已□□且正在□行。□可以使用 Host Utilities □件附□的 `sanlun` □用程序□□路径。例如，在 FC 多路径配置中，□可以使用 `sanlun fcp show adapter -v` 命令。

   在以下示例中，有□个 HBA 端口（ fcd0 和 fcd1 ）□接到□行正常的主机（端口状□）。□只能有一个 HBA 或 iSCSI □□程序，并且必□通□□目□ LUN 提供多个路径来配置多路径。

```
# sanlun fcp show adapter -v
adapter name: fcd0
WWPN: 50060b000038c428
WWNN: 50060b000038c429
driver name: fcd
model: A6826A
model description: Fibre Channel Mass Storage Adapter
(PCI/PCI-X)
serial number: Not Available
hardware version: 3
driver version: @(#) libfcd.a HP Fibre Channel
ISP 23xx & 24xx Driver B.11.23.04
/ux/core/isu/FCD/kern/src/common/wsio/fcd_init.c:Oct 18
2005,08:19:50
firmware version: 3.3.18
Number of ports: 1 of 2
port type: Fabric
port state: Operational
supported speed: 2 GBit/sec
negotiated speed: 2 GBit/sec
OS device name: /dev/fcd0
adapter name: fcd1
WWPN: 50060b000038c42a
WWNN: 50060b000038c42b
driver name: fcd
model: A6826A
model description: Fibre Channel Mass Storage Adapter
(PCI/PCI-X)
serial number: Not Available
hardware version: 3
driver version: @(#) libfcd.a HP Fibre Channel
ISP 23xx & 24xx Driver B.11.23.04
/ux/core/isu/FCD/kern/src/common/wsio/fcd_init.c:Oct 18
2005,08:19:50
firmware version: 3.3.18
Number of ports: 2 of 2 port type: Fabric
port state: Operational
supported speed: 2 GBit/sec
negotiated speed: 2 GBit/sec
OS device name: /dev/fcd1
```

如果在主机上□用了多路径，□可以□同一 LUN □示多个路径。□可以使用 `sanlun lun show all` 命令□□路径。在以下示例中，□可以□到指向同一 LUN 的多个路径（□： `/vol/vol1/lun` ）：

```
# sanlun lun show all
filer: lun-pathname device filename adapter protocol lun size      lun
state
fish: /vol/vol1/lun /dev/rdsk/c15t0d0 fcd0 FCP       10m (10485760) GOOD
fish: /vol/vol1/lun /dev/rdsk/c14t0d0 fcd1 FCP       10m (10485760) GOOD
```

3. 在 FC 配置中，使用交□机分区配置□主机 HBA 端口和目□端口□行分区。
4. 安装并□置相□的 FC 或 iSCSI 交□机。
5. □□ SnapDrive for UNIX 堆□要求。
6. 安装或升□□用于 UNIX 的 SnapDrive 。
7. □□ SnapDrive for UNIX 安装。
8. □到 `snapdrive.conf` 文件路径。
9. 在 `snapdrive.conf` 文件中配置以下配置□量：
   - ` 多路径□型 _`
   - `` default-transport_
   - ` fsttype_`
   - ` vmtype_`

   □于□台主机，多路径□型，□□□型，文件系□和卷管理器□型彼此相□。下表介□了所有可能的□合：

主机平台	默□□□□型	多路径□型	**FSType**	**vmtype**
Solaris	iSCSI （硬件）	无	UFS	无
iSCSI （□件）	MPxIO	UFS	无	无
UFS	Svm	FCP	DMP	VxFS

+ 上表提供了 ` multipathing-type_` ， ` default-transport_` ， ` fsttype_` 和 ` vmtype_` 配置□量支持的□。

10. □于 SFRAC □境，□行 SnapDrive config check cluster` 命令。
11. 保存 `snapdrive.conf` 文件。

    SnapDrive for UNIX 会在□次□□此文件□自□□其□行□□。要使更改生效，必□重新□□ SnapDrive for UNIX 守□□程。

    - 相□信息 *

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"《 Solaris Host Utilities 6.1 安装和□置指南》"

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